

SUPPLYCHAIN ATLAS

SupplyChain Atlas

Supply Chain Ontology and Governed Conversational Analytics

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Operational problem

Carrier updates, IoT sensor events, supplier reliability, plant demand, and customer commitments sit in separate systems.

Atlas identifies the shipment that should be escalated first and shows the business evidence behind that decision.

Data ontology and governance

SUPPLIERS, PLANTS, PARTS, PURCHASE_ORDERS, SHIPMENTS, SENSOR_EVENTS, CUSTOMER_ORDERS, and SHIPMENT_RISK form the ontology.

V_CONTROL_TOWER is the governed application surface. It joins evidence and prevents order-level double counting in the shipment queue.

Snowpark risk pipeline

Snowpark calculates: $\text{risk score} = \text{delay factor} + \text{sensor factor} + \text{supplier reliability factor}$.

Delay, temperature, shock, GPS status, and supplier tier remain inspectable inputs.

The deployment path is designed for Streams and Tasks to recalculate risk when new events arrive.

CoCo and semantic analytics

CoCo CLI can discover the catalog, validate the semantic model, generate and test SQL, and deploy Streamlit.

The semantic contract restricts conversational questions to trusted metrics such as at-risk revenue, critical shipments, supplier, plant, and risk level.

Live decision workflow

A late ETA or sensor anomaly refreshes a shipment score. V_CONTROL_TOWER connects it to affected customer revenue.

The Streamlit command center ranks disruptions, provides a Decision Receipt, and lets a manager simulate days recovered or alternate sourcing.

Live demo: 33 shipments, 5 critical disruptions, 3 suppliers, and \$40.2M revenue exposure.

The dashboard supports risk filters, shipment ranking, a recovery what-if lab, and an evidence receipt.